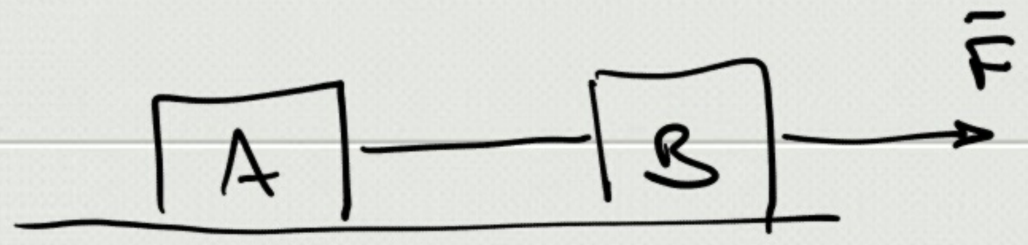
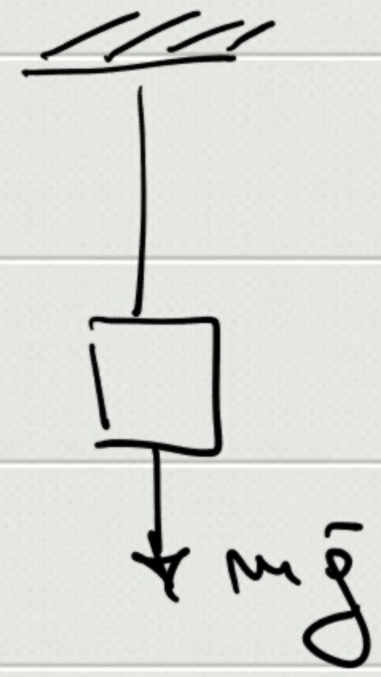
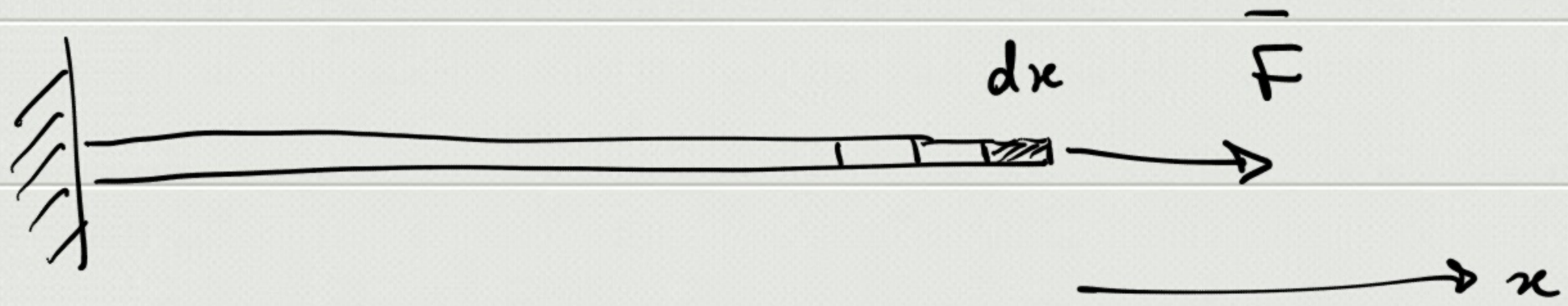
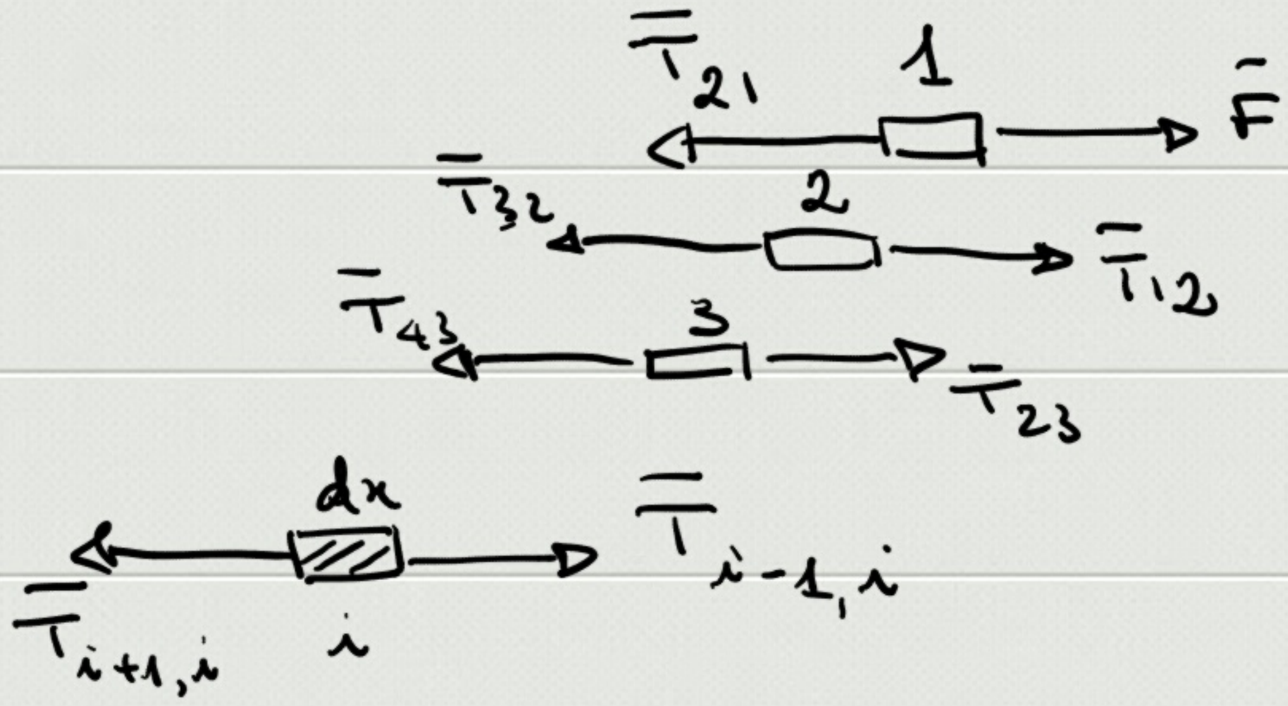




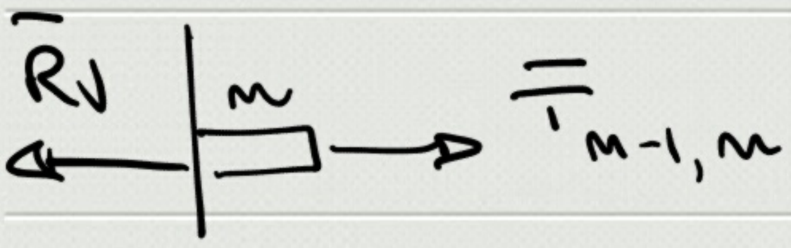
fili ideali : $m \simeq 0$, inestensibili



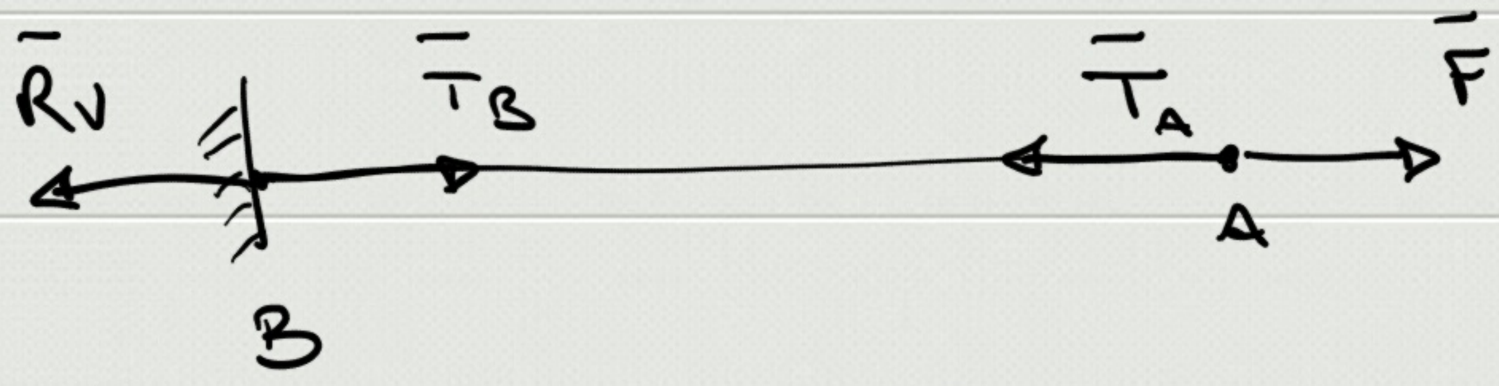


$\bar{T}_{21} = -\bar{F}$

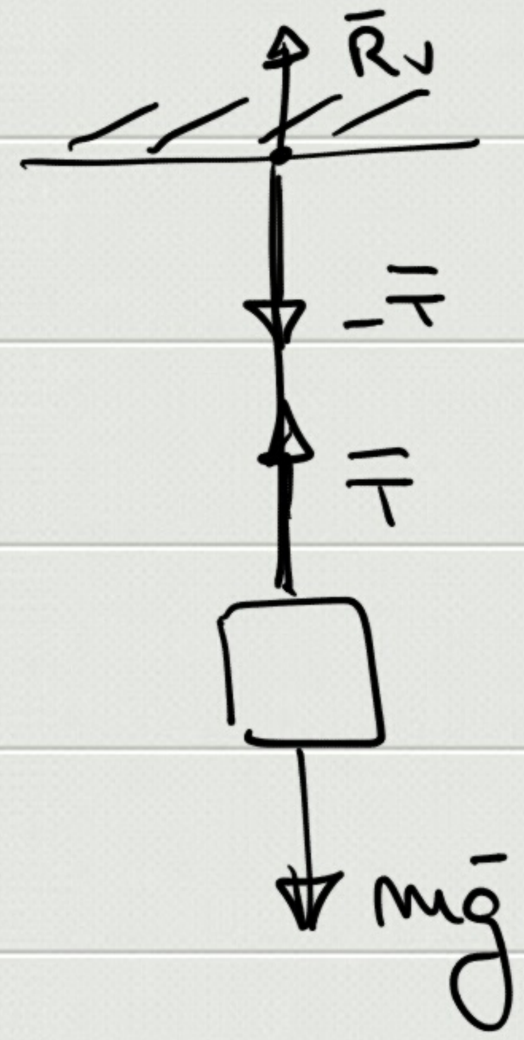
$\bar{T}_{12} = -T_{32}$



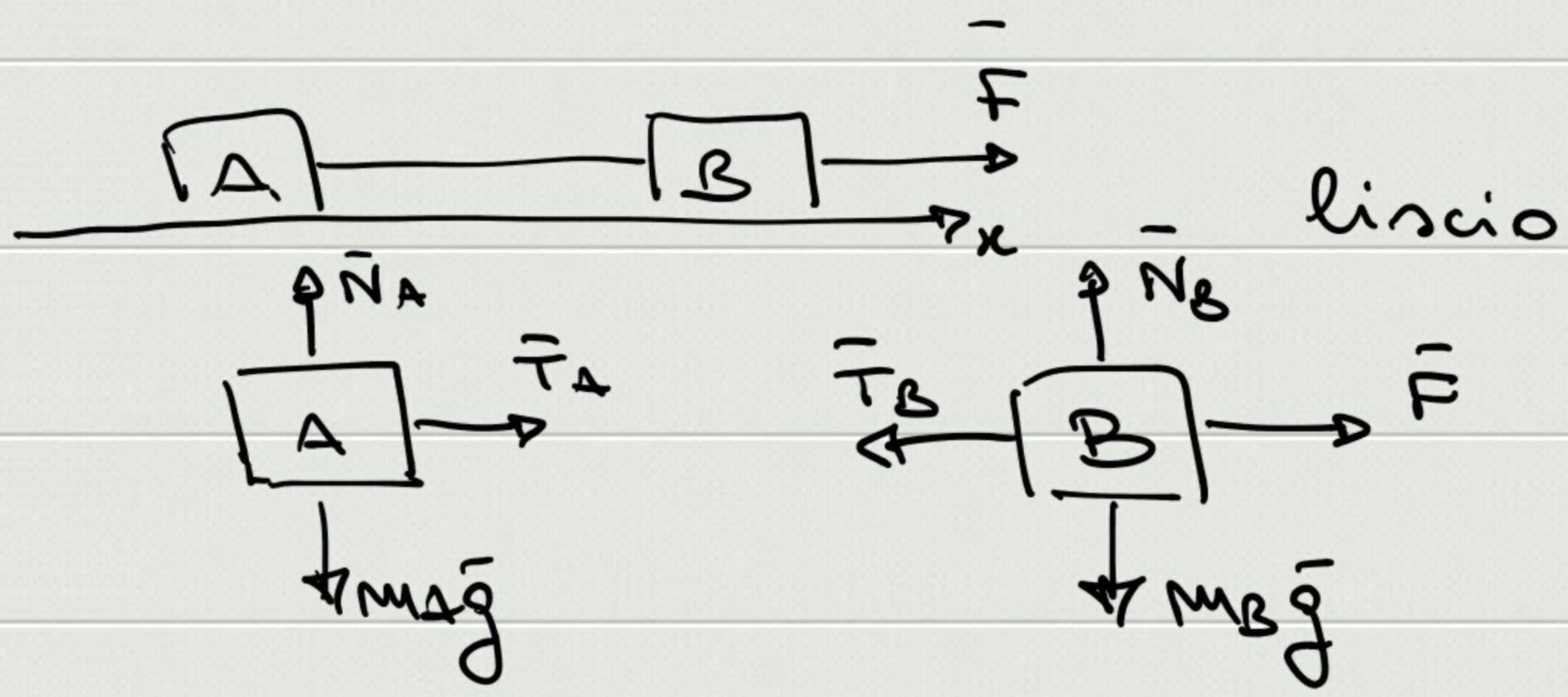
$\bar{R}_V = -\bar{T}_{m-1,m}$



$\bar{F} = -\bar{T}_A = \bar{T}_B = -\bar{R}_V$



$$\left.\begin{aligned} T &= R_v \\ T &= mg \end{aligned}\right\} \quad \boxed{mg = R_v}$$



$$\sum_i \bar{F}_i = m\bar{a}$$

$$T_A = T_B = T$$

$$a_A = a_B = a$$

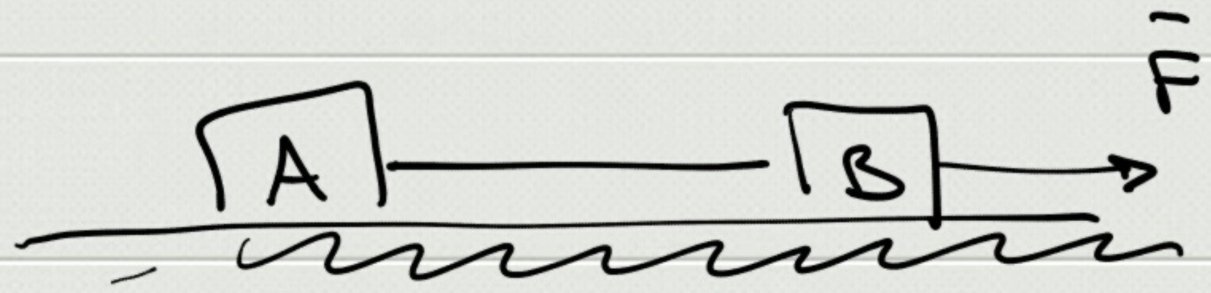
$$\begin{aligned} A&: \quad T_A = m_A a_A \\ B&: \quad -T_B + F = m_B a_B \end{aligned}$$

$$\begin{cases} T = m_A a \\ -T + F = m_B a \end{cases}$$

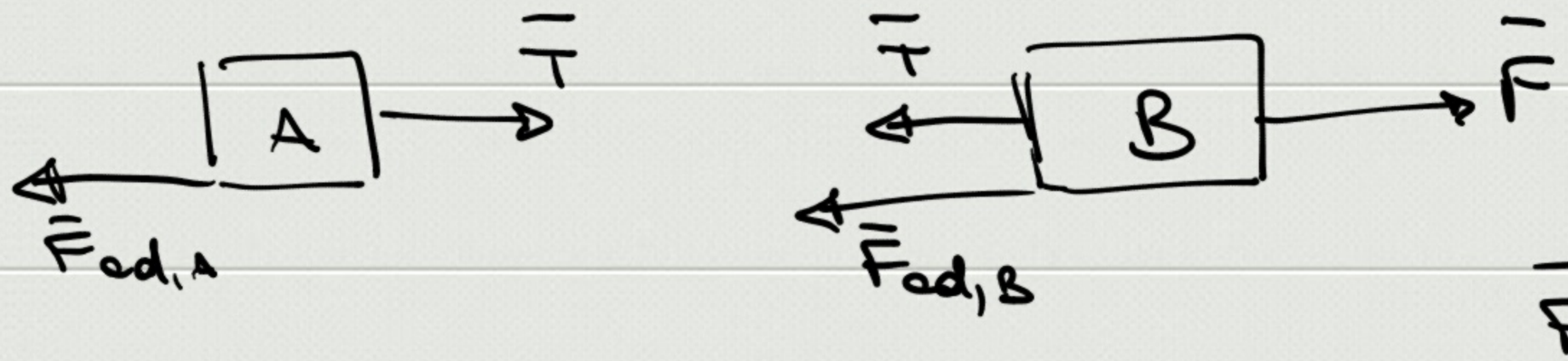
$$F = (m_A + m_B)a \quad \Rightarrow \quad a = \frac{F}{m_A + m_B}$$

$$F_A = T = m_A a = \frac{m_A}{m_A + m_B} F$$

$$F_B = F - T_B = m_B a = \frac{m_B}{m_A + m_B} F$$



μ_A, μ_B (att. dim.)



$$\bar{F}_{ad} = -\mu_d N \bar{v}_r$$

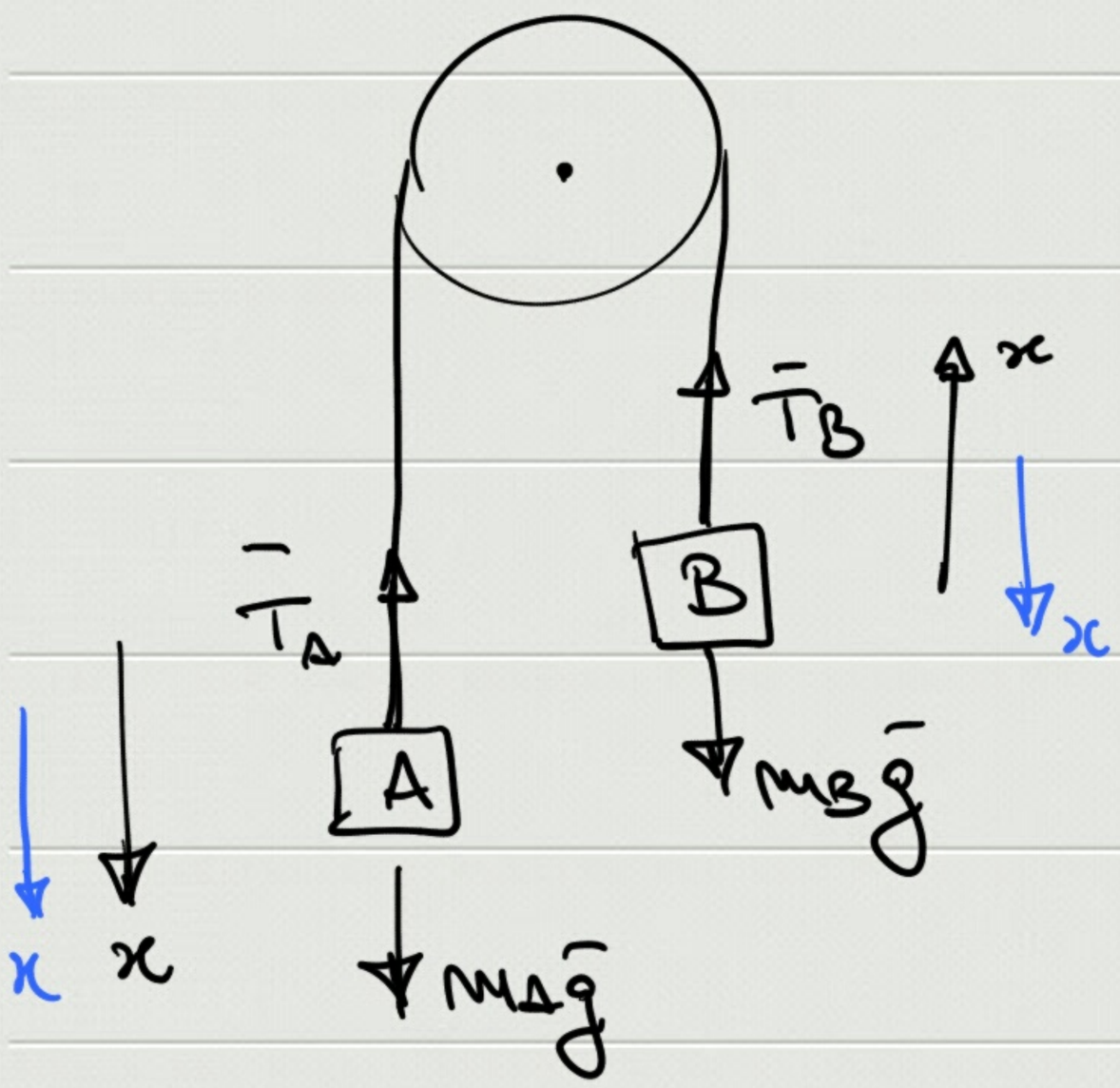
$$\begin{cases} A: & T - \mu_A m_A g = m_A a \\ B: & F - T - \mu_B m_B g = m_B a \end{cases}$$

$$F - (\mu_A m_A + \mu_B m_B) g = (m_A + m_B) a$$

$$\Rightarrow \quad a = \frac{F - (\mu_A m_A + \mu_B m_B) g}{m_A + m_B}$$

$$T = \frac{m_A}{m_A + m_B} \left[F + (\mu_A - \mu_B) m_B g \right]$$

Macchina di Atwood



$m_c \simeq 0$

$$A: \begin{cases} m_A g - T_A = m_A a_A \\ T_B - m_B g = m_B a_B \end{cases} \quad B:$$

$T_A = T_B$

$a_A = a_B = a$ *

$a_A = -a_B = a$

$$\begin{cases} m_A g - T = m_A a \\ T - m_B g = m_B a \end{cases}$$

$$(m_A - m_B) g = (m_A + m_B) a \implies \boxed{a = \frac{m_A - m_B}{m_A + m_B} g}$$